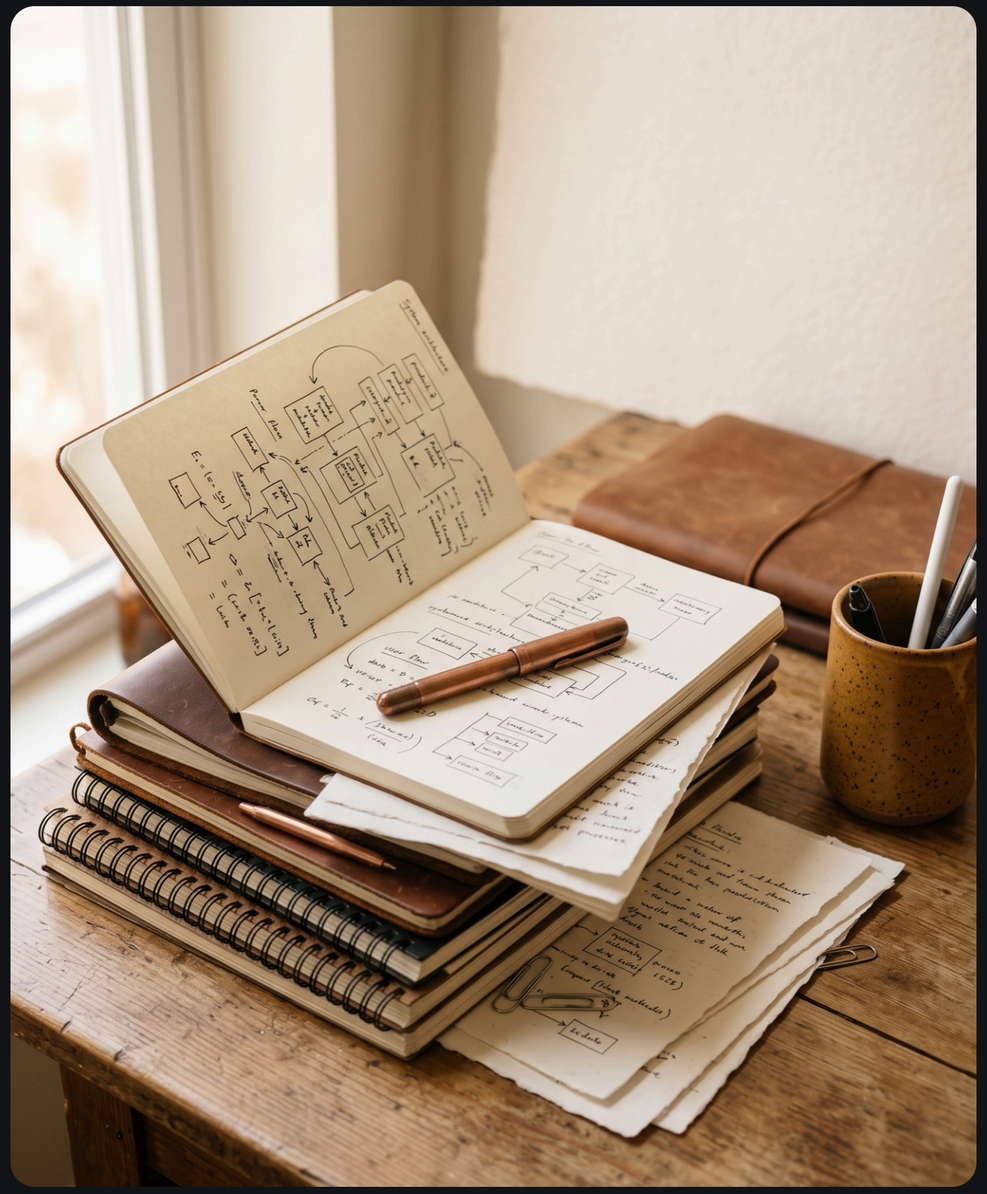USER GUIDE · LLM-WIKI

A wiki your LLM *keeps for you.*

Index and search your personal documentation with Karpathy's LLM Wiki pattern: compiled once, kept current, cross-linked.

BikS2013 · field notes
Athens · October 2026



BikS2013

Not retrieval at question time – *compilation* at ingest time.

01

You curate the sources

Drop articles, notes, PDFs and DOCX files into `raw/`. They are never modified – this is your source of truth.

02

The LLM writes the wiki

Every ingest produces a summary page and updates the entity and concept pages it touches, with wikilinks and citations.

03

You ask, it compounds

Search locally, ask with citations, file good answers back as pages, and lint for contradictions and gaps.

Pattern: Andrej Karpathy, "LLM Wiki" (gist, April 2026). Implementation: github.com/BikS2013/llm-wiki.



00 · THE IDEA

Three layers in one folder.

LAYER 1 · IMMUTABLE

Raw sources

Your documents, exactly as you added them. Read, never written — except the markdown conversion placed beside a PDF or DOCX.

```
raw/article.md
raw/report.pdf
raw/report.pdf.md
raw/assets/
```

LAYER 2 · LLM-OWNED

The wiki

Interlinked markdown pages the model creates and keeps consistent, plus a generated catalog and an append-only log.

```
wiki/index.md · wiki/log.md
wiki/sources/ · entities/
wiki/concepts/ · syntheses/
wiki/lint/
```

LAYER 3 · THE CONTRACT

The schema

A CLAUDE.md / AGENTS.md written by `init`: layout, page format, link rules and the ingest / query / lint workflows any agent must follow.

```
CLAUDE.md
AGENTS.md
.llm-wiki/ (tool state)
```

01

Set it up once.

Install the CLI, tell it where your wiki lives and which models to use, then create the folder layout. Nothing is defaulted silently — every missing setting is a clear error.



Install in three commands.

```
# requires Node 22+ and pnpm
git clone git@github.com:BikS2013/llm-wiki.git
cd llm-wiki
pnpm install
pnpm build            # tsc → dist/
npm link              # optional: puts `llm-wiki` on your PATH

llm-wiki --help
```

- **No PATH link?** Run it as `node dist/cli.js …` or, from the project, `pnpm dev -- …`.
- **Check the build** with `pnpm test`: 32 tests, no network needed.
- **Dependencies are vetted** – `pnpm audit` reports zero advisories at install time.

Configure once — four tiers, *no* hidden defaults.

1 · LOWEST
Shell env
What your shell already exports

2
~/.tool-agents/llm-wiki/.env
Your long-lived settings (0600)

3
./.env
Per-project overrides

4 · HIGHEST
CLI flags
--root, --provider, --model

SETTING	PURPOSE
`LLM_WIKI_ROOT`	Folder created by `init` · required by every command
`LLM_WIKI_PROVIDER / _MODEL`	Chat model for ingest, query, lint --llm
`LLM_WIKI_EMBEDDING_PROVIDER / _MODEL`	Optional · enables vector and hybrid search
`LLM_WIKI_PARSER_PROVIDER / _MODEL`	Optional · converts PDF and DOCX originals

```
# ~/.tool-agents/llm-wiki/.env
LLM_WIKI_ROOT=/Users/you/notes-wiki
LLM_WIKI_PROVIDER=anthropic
LLM_WIKI_MODEL=claude-opus-5
ANTHROPIC_API_KEY=…

LLM_WIKI_EMBEDDING_PROVIDER=openai
LLM_WIKI_EMBEDDING_MODEL=text-embedding-3-small
OPENAI_API_KEY=…

LLM_WIKI_PARSER_PROVIDER=gemini
LLM_WIKI_PARSER_MODEL=gemini-3.6-pro
GOOGLE_API_KEY=…
```

`llm-wiki config` shows every resolved setting, its tier, and masks the keys.

Eight providers, vendor-canonical variables.

PROVIDER	VARIABLES IT READS	CHAT	EMBED	PDF NATIVELY
openai	`OPENAI_API_KEY` · `OPENAI_BASE_URL` · `OPENAI_ORG_ID`	yes	yes	yes
anthropic	`ANTHROPIC_API_KEY` · `ANTHROPIC_BASE_URL`	yes	—	yes
gemini	`GOOGLE_API_KEY (or GEMINI_API_KEY)`	yes	yes	yes
azure-openai	`AZURE_OPENAI_API_KEY` · `_ENDPOINT` · `_DEPLOYMENT` · `_API_VERSION`	yes	yes	yes
azure-anthropic	`AZURE_AI_INFERENCE_KEY` · `AZURE_AI_INFERENCE_ENDPOINT`	yes	—	yes
ollama	`OLLAMA_HOST`	yes	yes	text extracted locally
litellm	`LITELLM_PROXY_URL` · `LITELLM_MASTER_KEY`	yes	yes	yes
mlx	`OPENAI_BASE_URL (OpenAI-compatible server)`	yes	yes	text extracted locally

Model names are yours to choose (e.g. claude-opus-5, gpt-5, gemini-3.6-flash, llama3). For azure-openai the chat model is the deployment in AZURE_OPENAI_DEPLOYMENT.

Create the wiki: *init*.

```
$ llm-wiki init ~/notes-wiki
initialised wiki at /Users/you/notes-wiki
  created …/raw
  created …/raw/assets
  created …/wiki/sources  entities  concepts  syntheses  lint
  created …/CLAUDE.md  AGENTS.md
  created …/wiki/index.md  wiki/log.md
next: export LLM_WIKI_ROOT=~/notes-wiki (or pass --root)

$ export LLM_WIKI_ROOT=~/notes-wiki
```

WHAT YOU GET

A folder that is also a repo

Plain markdown throughout. Put it under git for history; open it in Obsidian for the graph view; open it in Claude Code and the schema turns the agent into the wiki's maintainer.

SAFE TO RE-RUN

Existing files are kept

`init` refuses to overwrite a wiki that already has a schema; `--force` rewrites only CLAUDE.md and AGENTS.md.

02

The daily loop.

Ingest a source, search the pages, ask a question, file the answer, lint now and then. Each step leaves the wiki richer than it found it.



Ingest: one source in, *five to fifteen* pages touched.

```
$ llm-wiki ingest ~/Downloads/deploy-notes.md
reading related pages for raw/deploy-notes.md
asking anthropic/claude-opus-5 to compile raw/deploy-notes.md
# Deployment notes for the Ntakker API
Takeaways:
- Deploys to Azure Web App as a Docker container …
Pages written:
- replace wiki/sources/deploy-notes.md
- create wiki/entities/ntakker-api.md
- create wiki/entities/azure-blob-storage.md
- create wiki/concepts/configuration-management.md
  …

$ llm-wiki ingest                  # every pending source
$ llm-wiki ingest raw/x.md --dry-run
```

- **Context for the model:** the index plus the most related existing pages, found by search – so it updates pages instead of duplicating them.
- **After writing:** `index.md` is regenerated, the search index rebuilt, a log entry appended, and the source's hash recorded.
- **Idempotent:** an unchanged source is skipped; `--force` redoes it; `--dry-run` shows the plan only.
- **Never truncates:** a source above `--max-source-chars` is refused with a message, not silently cut.

Search: local, instant, *three* modes.

BM25 · DEFAULT

Keyword ranking

Okapi BM25 over heading-level chunks with a light stemmer. No model, no network, works the moment a page exists.

VECTOR

Semantic similarity

Cosine over embeddings cached by chunk hash. Run `llm-wiki index --embed` once; only new or changed chunks are embedded afterwards.

HYBRID

Both, fused

Reciprocal-rank fusion of the two lists. The default for `query` as soon as embeddings exist.

```
$ llm-wiki search "postgres docker"                    # bm25, wiki pages
$ llm-wiki search "how do I run the db" --mode hybrid --limit 5
$ llm-wiki search "psql" --layer all --json              # include raw/ sources, machine-readable
1. wiki/entities/postgres.md › Postgres  (0.727)
   Postgres runs in a Docker container on port 5432. See [[docker]] …
```

Query: answers with citations — *filed back* on request.

```
$ llm-wiki query "Where does the API read its config?" --file
### Where the API reads configuration
The [[entities/ntakker-api|Ntakker API]] fetches its configuration
at startup from [[entities/azure-blob-storage]] in the `app-config`
container ([[sources/deploy-notes]]).

### What remains undecided
Whether to move to Azure App Configuration ([[entities/giorgos]]).

### Sources
- [[sources/deploy-notes]] · [[entities/ntakker-api]] · …
filed as wiki/syntheses/where-does-the-api-read-its-config….md
```

- **Every claim cites a page.** If the wiki lacks the answer, the model says what is missing instead of inventing it.
- `--file` stores the answer under `wiki/syntheses/` with frontmatter, updates the index and logs it.
- **Tune the context:** `--limit`, `--pages`, `--context-chars`, `--mode`.

Lint: a health check in *two* passes.

STRUCTURAL · NO LLM

Deterministic findings

Dangling wikilinks · orphan pages · pages without outbound links · missing frontmatter or summary · basename collisions · stale `index.md` · sources in `raw/` not yet ingested · PDF/DOCX originals not yet converted.

SEMANTIC · --LLM

What only a reader can see

Contradictions between pages · claims superseded by newer sources · concepts mentioned repeatedly but lacking a page · missing cross-references · data gaps · suggested questions and sources.

```
$ llm-wiki lint                    # structural, exit 1 on errors
$ llm-wiki lint --fix              # also regenerate index.md and the search index
$ llm-wiki lint --llm --report     # semantic pass, filed as wiki/lint/<date>.md and logged
```

Two special files: *index.md* and *log.md*.

index.md — what is in the wiki

A catalog of every page grouped by type, each with a link, its one-line `summary`, its date and source count. Generated from page frontmatter by `llm-wiki index` — never hand-edit it; edit the frontmatter instead.

```
## Entities (6)
- [[azure-blob-storage|Azure Blob Storage]] — Object storage
  used for runtime configuration. _(2026-09-12, 1 source)_
- [[ntakker-api|Ntakker API]] — Backend API deployed to
  Azure Web App in Docker. _(2026-09-12, 1 source)_
```

log.md — what happened, when

Append-only. Every entry starts with the same prefix, so the log is grep-able: `grep "^## \[" log.md | tail -5`. The CLI reads it too: `llm-wiki log --last 5 --kind ingest`.

```
## [2026-09-12] ingest | Deployment notes for the Ntakker API
Source: raw/deploy-notes.md
Pages touched: … · Takeaways: …

## [2026-09-12] query | Where does the API read its config…
Filed as [[syntheses/where-does-the-api-…]].
```

`llm-wiki index --raw` also indexes raw sources for search; `--embed` computes missing embeddings.

03

Documents, originals intact.

A dedicated parser model converts the document to markdown. The original stays untouched in `raw/` and every derived page points back to it.



A separate *parser* model, chosen for the job.

1 · YOU ADD

raw/report.pdf

Any PDF or DOCX, copied into `raw/` if it was elsewhere. It is never modified.

→

2 · PARSER LLM

faithful markdown

Headings, lists, tables, figures as bracketed descriptions. Nothing summarised, nothing dropped. Set by `LLM_WIKI_PARSER_PROVIDER / _MODEL`.

→

3 · THE WIKI READS

raw/report.pdf.md

Written beside the original with provenance frontmatter, then ingested like any other source.

```
$ llm-wiki ingest raw/report.pdf            # convert, then ingest
$ llm-wiki parse raw/contract.docx          # convert only — review raw/contract.docx.md first
$ llm-wiki ingest raw/report.pdf --reparse  # redo the conversion; cached by hash otherwise
```

Provenance travels with every page.

```
# raw/report.pdf.md
---
title: Quarterly Report
original: raw/report.pdf
original_type: PDF
original_sha256: 3f2a9c…
parsed_by: gemini/gemini-3.6-pro
parsed_at: 2026-09-12
---
> **Converted from the original PDF document
> [[raw/report.pdf]]** (412 KB) by gemini/gemini-3.6-pro
> on 2026-09-12. The original is kept unmodified in raw/.

# Quarterly Report
…
```

ON THE WIKI PAGE TOO

sources/quarterly-report.md

Gets `original: raw/report.pdf` in its frontmatter and a link to the original in its body – stamped by the tool, whatever the model wrote.

NATIVE VS EXTRACTED

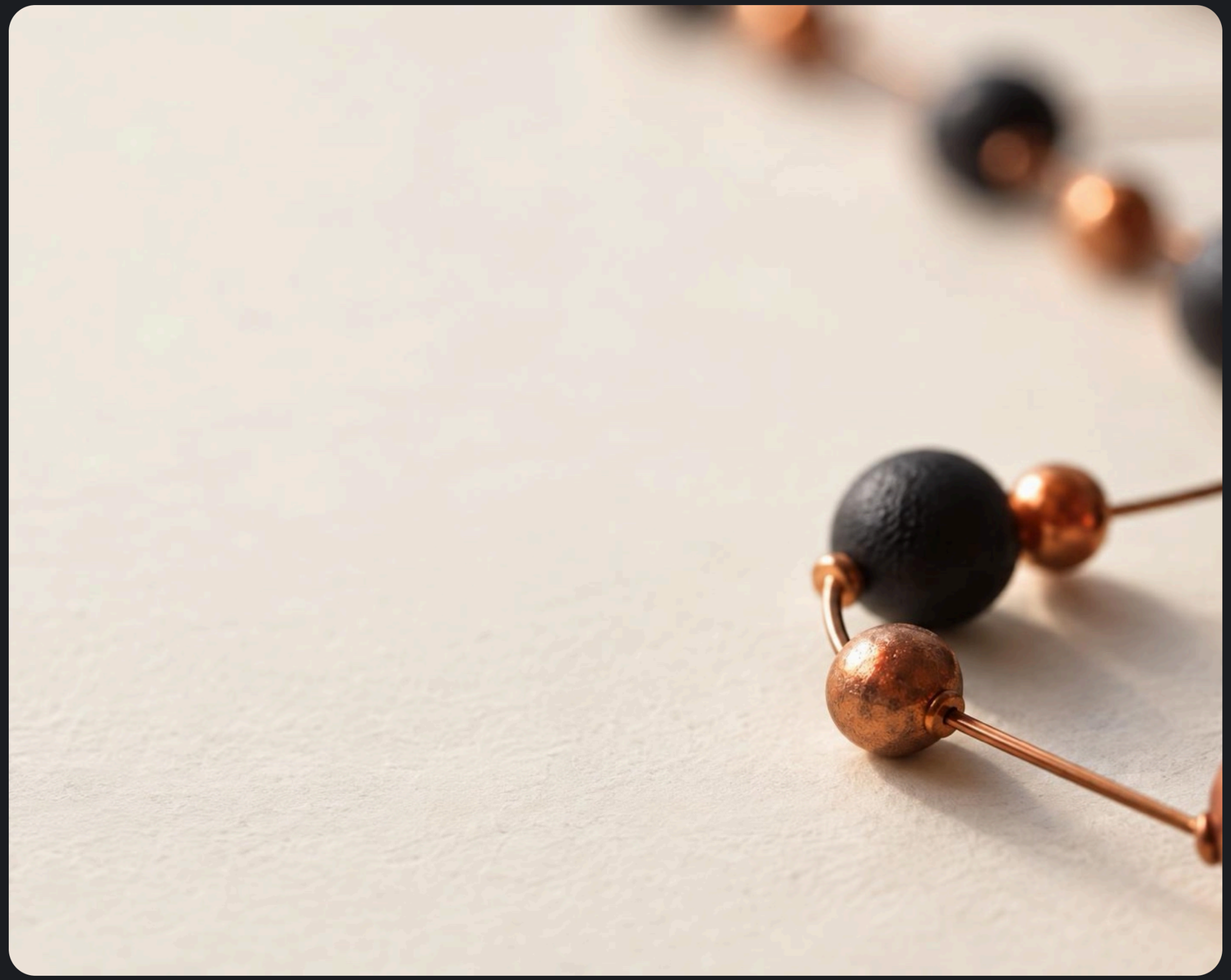
Scanned PDFs work

Anthropic, Gemini, OpenAI, Azure and LiteLLM receive the PDF itself. Ollama and MLX receive text extracted with pdf.js – a scanned page yields nothing there, and the tool says so.

04

Let the agent do the bookkeeping.

Obsidian is the IDE, the LLM is the programmer, the wiki is the codebase. The schema and the MCP server make that literal.



The schema turns any coding agent into the *maintainer*.

§1

Layout & page format

Where each page type lives, the frontmatter keys, kebab-case slugs, wikilink conventions, citing `raw/` for every non-obvious claim.

§2

Operations

Step-by-step ingest, query and lint workflows — including "run `llm-wiki index` after editing pages" and how to flag a contradiction.

§3

Working style

Never touch `raw/`; keep pages atomic; prefer updating over duplicating; mark possible duplicates for the next lint pass.

```
$ cd ~/notes-wiki && claude          # CLAUDE.md is read automatically; AGENTS.md serves Codex and others
> ingest raw/podcast-notes.md and tell me what changed in the overview
```

The MCP server: native tools for the agent.

```
$ claude mcp add llm-wiki -- \
    node /path/to/llm-wiki/dist/cli.js \
    --root ~/notes-wiki mcp

# or run it yourself for any MCP client
$ llm-wiki mcp                # stdio
```

Settings resolve the same way as for the CLI, so the server needs no extra configuration beyond `LLM_WIKI_ROOT` (or `--root`).

TOOL	DOES
`wiki_search`	bm25 / vector / hybrid, any layer
`wiki_read_page`	a page, or index / log
`wiki_list_pages`	id, type, title, summary; filter by type
`wiki_write_page`	create / replace / append; reindexes
`wiki_append_log`	an entry in the standard format
`wiki_recent_log`	the last N entries
`wiki_lint`	the structural health check
`wiki_reindex`	index.md + search index

What a page looks like.

```
---
title: Ada Lovelace
type: entity          # entity | concept | source | synthesis
summary: One sentence shown verbatim in index.md.
tags: [mathematics, computing]
created: 2026-04-02
updated: 2026-04-09
sources: [raw/some-article.md]
aliases: [Countess of Lovelace]
---
# Ada Lovelace

Worked with [[charles-babbage]] on the
[[concepts/analytical-engine|Analytical Engine]]
([[sources/some-article]]).

> **Contradiction:** [[sources/a]] dates the notes to 1842,
> [[sources/b]] to 1843.
```

- **Frontmatter feeds the index:** `title`, `summary`, `type`, `updated`, `sources`.
- **Wikilinks** by basename when unique, by `folder/slug` when not; lint flags collisions and dangling links.
- **Contradictions are callouts**, never silent overwrites.
- **After hand edits**, run `llm-wiki index`.

A worked example, from one small note.

8

pages from a four-line deployment note

One source page, six entities (the API, the team member, three Azure services, Docker) and one concept – all cross-linked, all indexed.

STEP	RESULT
ingest	8 pages written · index and log updated · 17 chunks indexed
index --embed	17 embeddings computed once, cached by hash
search --mode hybrid	"where does production config live" → the configuration-management page, first hit
query --file	Answer cited 5 pages · filed under `syntheses/`
lint --llm --report	Suggested a missing page (Azure App Configuration), one missing cross-reference, three next questions

Run on 2026-09-12 with gemini-3.6-flash as chat and parser model and text-embedding-3-small for embeddings.

When it stops, it tells you why.

MESSAGE	WHAT TO DO
`Missing required configuration setting LLM_WIKI_ROOT`	Export it, add it to `~/.tool-agents/llm-wiki/.env`, or pass `--root`. Same for any `LLM_WIKI_*` or provider key.
`… is not an initialised wiki (missing raw/)`	Run `llm-wiki init <dir>` first.
`… is a PDF/DOCX original and needs the parser LLM`	Set `LLM_WIKI_PARSER_PROVIDER` and `_MODEL`.
`Search mode "hybrid" needs an embedding provider / No embeddings are stored yet`	Set the embedding settings, then `llm-wiki index --embed`.
`… hit the N-token output limit before finishing`	Raise `--max-tokens` (ingest, query, lint) or `--parse-max-tokens`.
`… is above the --max-source-chars / --max-bytes limit`	Split the source or raise the limit – the tool never truncates silently.
`This model … is no longer available`	A provider retired the model name; pick a current one with `--model`.

Add `--verbose` to any command for a stack trace. Exit codes: 2 configuration, 3 LLM request, 1 anything else.

CHEAT SHEET

Your documentation, *compiled.*

```
llm-wiki init ~/notes-wiki  &&  export LLM_WIKI_ROOT=~/notes-wiki
llm-wiki ingest raw/<file>            # markdown, text, or PDF / DOCX via the parser model
llm-wiki search "<terms>"              # bm25 · --mode hybrid after `index --embed`
llm-wiki query "<question>" --file     # cited answer, filed as a synthesis page
llm-wiki lint --llm --report           # contradictions, gaps, next questions
llm-wiki log --last 5                  # what happened lately
llm-wiki mcp                           # give your agent the tools
```

BikS2013

github.com/BikS2013/llm-wiki
Pattern: Karpathy, LLM Wiki, 2026